

School of Arts and Sciences

Department of Computer Science and Information Technology

Fall 2025 – 2026

CSCI392 Computer Networks

Class Day	:	_____	Class Time:	_____	Room:	_____
Instructor	:	_____				
E-mail	:	_____				
Office	:	_____	Office Hours:	_____	Extension No:	_____

I. Course Description

The **CCNA** course series describes the architecture, components, and operations of routers and switches in a small network. Students learn how to configure a router and a switch for basic functionality.

II. Course Description

Students will get deeper understanding, in continuation to previous introductory courses, into how network connectivity is made possible through intermediary network elements, namely the routers and switches, and their main functionalities at the theoretical plane. Basic network security and monitoring concepts are additionally introduced at the end of this course. At the practical plane, students learn how to configure a router and a switch for basic functionality, and how to troubleshoot and resolve common network issues with static routing, Virtual Local Area Networks, LAN security configuration, mitigating common network devices attacks, Dynamic host configuration, Gateway redundancy.

The CCNA® Switching, Routing, and Wireless Essentials curriculum Version 7.0 helps students enrolled in the Cisco Networking Academy® program prepare for entry-level career opportunities, continuing education with networking related degree programs, and globally recognized Cisco certifications.

The 3-course CCNA® series includes three courses that make up the recommended course flow: Introduction to Networks, Switching, Routing, and Wireless Essentials, and Enterprise Networking, Security, and Automation.

LIU follows the above course flow as recommended by Cisco in order to help students prepare for the CCENT certification exam after the first two courses, and to prepare for the CCNA Routing and Switching certification exam after completing all four courses.

The course matches mainly with the second course in the flow, Routing and Switching Essentials. By the end of this course, students will be able to configure and troubleshoot routers and switches and resolve common issues using virtual LANs, inter-VLAN routing, DHCP configuration and protect the network from the common network attacks.

III. Course Learning Outcomes

A student who successfully fulfills the course requirements will have demonstrated ability to:

Description	
CSCI392 Outcomes	CO – 1 Analyze, Examine, Apply configurations and Test Layer 3 connectivity in a statically routed network topology (static routing approach) and in a dynamically routed network topology (dynamic routing protocols)
	CO – 2 Analyze, Examine, Apply configurations and Test Layer 2 connectivity in a switched network topology (Ethernet Switch MAC address table, Port settings, and VLAN Implementation)
	CO – 3 Analyze and Identify how VLANs create logically separate networks and layer 2 and layer 3 connectivity planes can interrelate (inter-VLAN routing)
	CO – 4 Interpret the network security policies on Layer 2 by restricting access to specific number of devices on each switch port.
	CO – 5 Understand the common LAN attacks and apply configurations that aim to protect the LAN network from these attacks. Understand DHCP protocol and configure a router to do this operation in a small network.

IV. CSCI392 Outcomes

A student who successfully fulfills the course requirements will have demonstrated ability to:

		Description
CSCI392 Outcomes	CO-1	Understand, Apply configurations and Explain about Layer 3 connectivity in a statically routed network topology (static routing approach) and in a dynamically routed network topology (dynamic routing protocols)
	CO-2	Understand, Apply configurations and Explain about Layer 2 connectivity in a switched network topology (Ethernet Switch MAC address table, Port settings, and VLAN implementation)
	CO-3	Understand and Explain how VLANs create logically separate networks and layer 2 and layer 3 connectivity planes can interrelate (inter-VLAN routing)
	CO-4	Interpret the network security policies on Layer 2 by restricting access to specific number of devices on each switch port.
	CO-5	Understand the common LAN attacks and apply configurations that aim to protect the LAN network from these attacks. Understand DHCP protocol and configure a router to do this operation in a small network.

CSCI392 Outcomes mapping to exams (an '✓' to indicate the coverage)

		Midterm	Skills	Final Online	Final
CSCI392 Outcomes	CO – 1	✓	✓	✓	✓
	CO – 2	✓	✓	✓	
	CO – 3	✓	✓	✓	✓
	CO – 4		✓	✓	✓
	CO – 5		✓	✓	✓

CSCI392 Outcomes mapped to Midterm

		Midterm				
		Part 1	Part 2		Part 3	
		Q1 (20 pts)	Q1 (25 pts)	Q2 (15 pts)	Q1 (20 pts)	Q2 (20 pts)
CSCI 392 Outcomes	CO – 1	05 pts			20 pts	20 pts
	CO – 2	05 pts		15 pts		
	CO – 3	10 pts	25 pts			
	CO – 4					
	CO – 5					

CSCI392 Outcomes mapped to the Final Exam

		Final				
		Part 1	Part 2		Part 3	
		Q1 (20 pts)	Q1 (20 pts)	Q2 (20 pts)	Q1 (10 pts)	Q2 (30 pts)
CSCI392 Outcomes	CO – 1				10 pts	
	CO – 2					
	CO – 3		15 pts			
	CO – 4	15 pts				30 pts
	CO – 5					

V. Topics Covered

1. *Routing Concepts*
2. *Static Routing*
3. *Switched Networks*
4. *LAN security concepts*
5. *Switch security Configuration*
6. *Virtual Networks VLANs*
7. *DHCP configuration on routers*

VI. Prerequisite – Corequisite

CSCI250 Introduction to Programming
 CSCI342 Fundamentals of Networking
 Technologies Or
 CCNA100 Introduction to Networks

Students are advised to take CCNA Certification [200-120 CCNA](#)

VII. Course Material: Book or handouts**Textbook(s)**

Routing and Switching Essentials Companion Guide.

Publisher: Cisco Press; 1 edition (Feb 18, 2014)

ISBN-10: 1-58713-318-0

Routing and Switching Essentials (Lab Manual)

By Cisco Networking Academy. **Publisher:** Cisco Press; 1 edition (Oct 17, 2013)

ISBN-10: 1-58713-320-2

VIII. Software required (for computer-lab courses)

Cisco Packet Tracer v8.2

IX. Detailed Schedule

Week		Book Reference	Topic Details
Week 1	Lecture 1	Course Introduction	Course Introduction - Explaining the course outline
	Lecture 2	Basic Device Configuration	Practice on packet tracer: - Create a topology and connect devices
Week 2	Lecture 1	Basic Device Configuration	Practice on packet tracer: - Configure a Switch with Initial Settings
	Lecture 2	Basic Device Configuration	Practice on packet tracer: - Configure a router with Initial Settings
Week 3	Lecture 1	Basic Device Configuration	- Configure Switch Ports - Secure Remote Access - Basic Router Configuration - Verify Directly Connected Networks
	Lecture 2	Basic Device Configuration	Practice: - 1.3.6 – Packet Tracer – Configure SSH - 1.4.7 – Packet Tracer – Configure Router

			Interfaces
Week 4	Lecture 1	Routing Concepts	- Path Determination - Packet Forwarding - Basic Router Configuration Review
	Lecture 2	Routing Concepts	- IP Routing Table - Static and Dynamic Routing
Week 5	Lecture 1	Routing Concepts	Practices: - 14.3.5 – Basic Router Configuration Review
	Lecture 2	IP Static Routing	- Static Routes - Configure IP Static Routes - Configure IP Default Static Routes - Configure Floating Static Routes - Configure Static Host Routes
Week 6	Lecture 1	IP Static Routing	Practice: - 15.6.1 – Configure IPv4 and IPv6 Static and Default Routes
	Lecture 2	Switching Concepts	- Frame Forwarding - Switching Domains
Week 7	Lecture 1	Switching Concepts	- Continue Module 2
	Lecture 2		
Week 8	Lecture 1	Practical exam on packet tracer during class time	
	Lecture 2	VLANs	- Overview of VLANs - VLANs in a Multi-Switched Environment - VLAN Configuration
Week 9	Lecture 1	VLANs	- VLAN Trunks - Dynamic Trunking Protocol
	Lecture 2	revision	
- MIDTERM EXAM on Friday 28 November from 8:30-9:45 am			
Week 10	Lecture 1	VLANs	Practices: - 3.3.12 – VLAN Configuration
	Lecture 2	VLANs	Practices: - 3.4.6 – Configure VLANs and Trunks - 3.5.5 – Configure DTP
Week 11	Lecture 1	Inter-VLAN Routing	- Inter-VLAN Routing Operation - Router-on-a-Stick Inter-VLAN Routing - Inter-VLAN Routing using Layer 3 Switches (overview) - Troubleshoot Inter-VLAN Routing
	Lecture 2	Inter-VLAN Routing	Practices: - 4.2.7 – Configure Router-on-a-Stick Inter-VLAN Routing
Week 12	Lecture 1	DHCPv4	- DHCP4 Concepts - Configure a Cisco IOS DHCP4 Server
	Lecture 2	DHCPv4	- Configure a Cisco IOS DHCP4 client
Week 13	Lecture 1	DHCPv4	- 7.2.10 Packet Tracer - Configure DHCPv4 - 7.4.1 Packet Tracer - Implement DHCPv4

(Wed and Thur holiday)	Lecture 2	LAN Security Concepts	Overview of these concepts: - Endpoint Security - Access Control
Week 14 (Holiday)	Lecture 1	LAN Security Concepts	- Layer 2 Security Threats - MAC Address Table Attack - LAN Attacks
	Lecture 2	Chapter 8 - Switch Security Configuration	- Implement Port Security - Mitigate VLAN Attacks - Mitigate DHCP Attacks - Mitigate ARP Attacks
Week 15 (Tue holiday)	Lecture 1	Chapter 8 - Switch Security Configuration	Practices: - 11.1.10 – Implement Port Security - 11.6.1 Packet Tracer - Switch Security Configuration
	Lecture 2	Skills Exam	During class time

Grading Policy and Withdrawal Deadline

Exams	LIU Midterm Exam	35%	(Date and time is decided later)
	LIU Final Exam	35%	(Date and time is decided later)
	Practical exam	10%	Week 8 During Class Time
	Skills Exam	20%	Week 15 During Class Time

Withdrawal Deadline: Week 5

Instructions for Exams

- Time allowed: **75 minutes for Midterm** whereas **90 minutes for Final Exam**.
- Cheating in any way, form or method will be rewarded with an F grade.
- You should read each question carefully before answering.
- You should solve the questions that you are confident about their answers first.
- Neither an Open Book nor an Open Notes exam.
- Laptop and USB not allowed

X. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	B+
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	